Exercises 3.10

19, 25, 33, 35, 40 (20%).

19–22 (a) Find the differential dy and (b) evaluate dy for the given values of x and dx.

19. $y = e^{x/10}, \quad x = 0, \quad dx = 0.1$

$dy = y' \, dx$

$y' = e^{\frac{x}{10}} \cdot \frac{1}{10}$

$y'(0) = e^0 \times \frac{1}{10} = \frac{1}{10}$ < $dy = \frac{1}{10} e^{\frac{x}{10}} \cdot dx$

$dy = \frac{1}{10} \times 0.1 = 0.01$

25. $y = \sqrt{x-2}$

$\Delta y = y(x+\Delta x) - y(x)$

$= \sqrt{3.8-2} - \sqrt{3-2}$

$= 0.34$

$dy = f'(x)\,dx$

$= [(x-2)^{\frac{1}{2}}]' \cdot 0.8$

$= \frac{1}{2}(x-2)^{-\frac{1}{2}} \times 1 \times 0.8$

$= \frac{1}{2}(3-2)^{-\frac{1}{2}} \times 0.8$

$= 0.4$

$f(x) = \sqrt[3]{x}$ $\quad f'(x) = \frac{1}{3}x^{-\frac{2}{3}}$

33. $L(x) = f(a) + f'(a)(x-a)$

$L(1001) = f(1000) + f'(1000)(x-1000)$

$= \sqrt[3]{1000} + \frac{1}{3}(1000)^{-\frac{2}{3}}(1001-1000)$

$= 10 + 3.33 \times 10^{-3}$

$= 10.00333$

34. $f(x) = e^x$ $\quad f'(x) = e^x$

$L(0.1) = f(0) + f'(0)(x-0)$

$= e^0 + e^0(0.1-0)$

$= 1 + 0.1 = 0.9$ $\quad 1.01$

40. Let $f(x) = (x - 1)^2$ $g(x) = e^{-2x}$

and $h(x) = 1 + \ln(1 - 2x)$

(a) Find the linearizations of f, g, and h at $a = 0$. What do you notice? How do you explain what happened?

(b) Graph f, g, and h and their linear approximations. For which function is the linear approximation best? For which is it worst? Explain.

a) $f'(x) = 2(x-1)$ $g'(x) = e^{-2x} \cdot -2$

$L_f(x) = f(a) + f'(a)(x-a)$

$= (0-1)^2 + 2(0-1)(x-0)$

$= 1 - 2x$

$L_g(x) = g(a) + g'(x)(x-a)$

$= e^{-2\cdot 0} - 2e^0 (x-0) = 1 - 2x$

$h'(x) = \frac{1}{1-2x} \cdot (-2)$

$L_h(x) = h(a) + h'(a)(x-a)$

$= 1 + \ln(1-0) - \frac{2}{1-0}(x-0)$

$= 1 + 0 - 2x = 1 - 2x$